

No effect of fish oil supplementation on serum inflammatory markers and their interrelationships: a randomized controlled trial in healthy, middle-...

BACKGROUND: A high intake of n-3 polyunsaturated fatty acids (PUFAs), mainly present in fish, may be associated with decreased inflammation. Previous intervention studies on fish PUFA and inflammatory markers in healthy individuals did not analyze a broad spectrum of inflammatory cytokines, chemokines and cell adhesion molecules, or their interrelationships. Therefore, we determined the effects of fish oil supplementation on 19 serum inflammatory markers and their interrelationships in healthy, middle-aged individuals. **METHODS:** Individuals (n=77) aged 50-70 years completed a randomized, double-blind placebo-controlled intervention study. Participants received 3.5 g/day fish oil (1.5 g/day total n-3 PUFA) (n=39) or placebo (high oleic sunflower oil) (n=38) for 12 weeks. Serum concentrations of 19 inflammatory markers were determined using a multiplex immunoassay before and after intervention. Changes in concentrations were analyzed using analysis of covariance and differences in patterns in inflammatory markers between the fish oil and placebo group were analyzed by principal component analysis. **RESULTS:** Fish oil supplementation did not significantly affect serum concentrations of cytokines, chemokines or cell adhesion molecules as compared with placebo. However, there was a trend for all inflammatory markers to increase after fish oil supplementation. PCA did not result in markedly distinctive patterns of inflammatory markers for the fish oil and placebo group. **CONCLUSION:** In conclusion, this 12-week randomized, double-blind placebo-controlled intervention trial did not show that 1.5 g/day n-3 PUFA significantly affected the serum inflammatory response in healthy individuals, nor did patterns of inflammatory markers. Thus, a healthy middle-aged population may not benefit from fish oil as an anti-inflammatory agent.